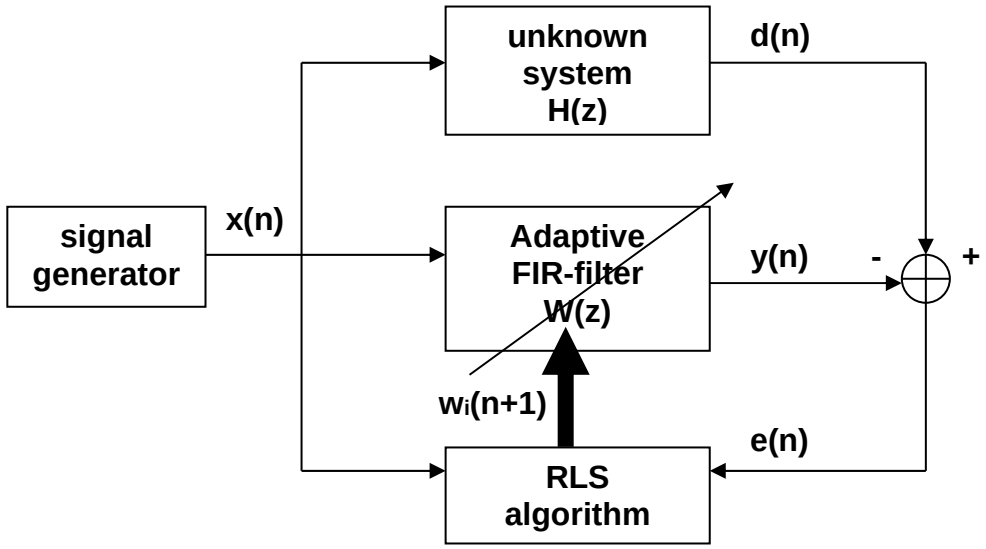
Problem 12.1 (MATLAB)
The RLS algorithm and Adaptive System Identification

This problem is a continuation of Problem 11.1.
Here adaptive system identification will be studied again, now using the RLS algorithm as shown below:

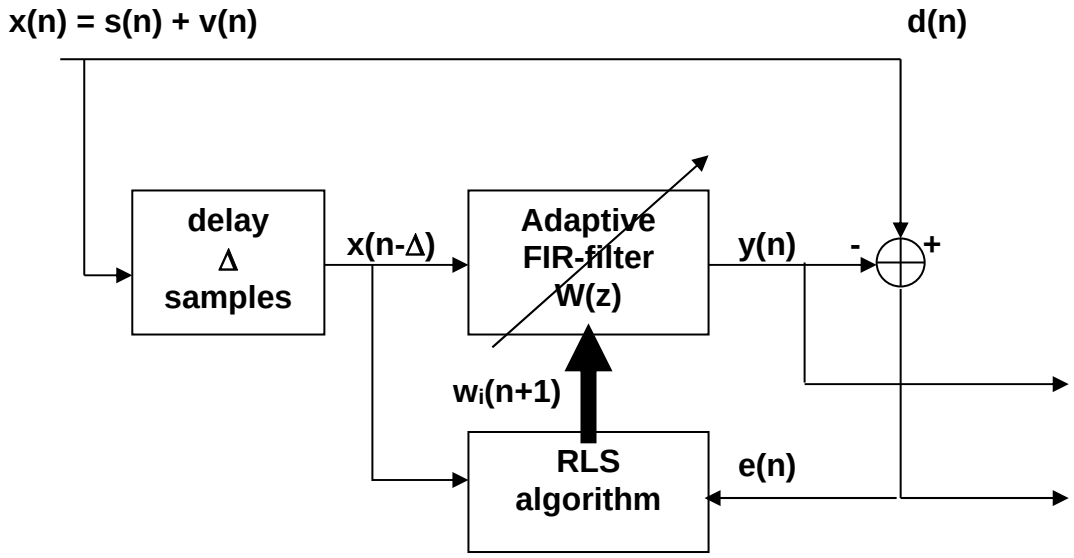


System set-up and experiments

- Use again the system set-up from Problem 11.1
- Implement the standard RLS algorithm as a function m-file or use the implementation made available accompanying this problem
- Compare the performance of the RLS algorithm in terms of convergence speed, stationary error and achieved estimate of **H(f)** with the performance of the LMS/LNLMS algorithm for some of the experiments from Problem 11.1

Problem 12.2 (MATLAB)
The RLS algorithm and Adaptive Linear Prediction / Noise Reduction

This problem is a continuation of Problem 11.2.
Here adaptive linear prediction will be studied again, now using the RLS algorithm as shown below:



System set-up and experiments

- Use again the system set-up from Problem 11.2
- Compare the performance of the RLS algorithm in terms of convergence speed, stationary error and achieved SNR improvement with the performance of the LMS/LNLMS algorithm for some of the experiments from Problem 11.2